

Battery Management System (BMS) Modelling and State of Charge (SOC) Estimation in MATLAB

Project synopsis submitted in partial fulfillment

For the Award of

CERTIFICATION

In

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By

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CHAPTER 1

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CHAPTER 2

PROJECT DESCRIPTION

This project's main goal is to create a comprehensive Battery Management System (BMS) model using MATLAB. This BMS model should be able to precisely incorporate important variables including capacity characteristics, voltage, and current when determining the State of Charge (SOC) of a battery pack during its charge and discharge cycles.

We must translate the battery's behavior into a mathematical representation in order to achieve this. This mathematical model should incorporate internal resistance, discharge capacity curves, and voltage-current characteristics. In essence, it should allow us to simulate how the battery would behave under various loads and charge/discharge scenarios

We must also develop techniques for estimating SOC. These algorithms will employ readily available metrics, like as voltage and current readings, to determine the SOC. In this case, different SOC estimating methods, including Equivalent Circuit Models, Extended Kalman Filtering, and Coulomb counting, can be studied and their accuracy assessed.

Collecting information will be necessary to confirm the BMS model's performance. This can be done by combining the MATLAB model with data collection tools or by utilizing experimental data that has previously been collected. Depending on the project's constraints and available resources, we can either use preexisting datasets or collect data from real battery cells.

Ultimately, in order to gather data and assess the SOC estimation performance over time, graphical visualizations must be created. These statistics must clearly show the true SOC values in proportion to the estimated SOC values. This example will allow us to assess the correctness of our BMS model and compare various SOC calculating techniques.

CHAPTER 3

Battery Management System (BMS) Modeling and State of Charge (SOC) Estimation in MATLAB

SOC Estimation

An important indicator of a battery's performance is its State of Charge (SOC). It shows the battery's remaining capacity as a proportion of its overall capacity. Since batteries are sealed chemical energy storage devices, the SOC cannot be tested directly in the context of battery technology. Rather, quantifiable battery parameters like voltage and current must be used to evaluate it.

The SOC of a battery is typically expressed as follows:

$$\text{SOC}(t) = \text{SOC initial} - \int_0^t \eta i(t) dt / \text{Crated}$$

In this equation, $\text{SOC}(t)$ represents the SOC at a given time, SOC initial is the initial SOC, Crated is the rated capacity of the battery, η is the charge- discharge efficiency, and $i(t)$ is the battery current.

In real-world applications, precise SOC data is essential, particularly when estimating battery life. Because excessive charging or discharging can cause irreversible internal battery damage, Battery Management Systems (BMS) use this information to decide when to cease charging and discharging.

Different SOC Estimation Methods

Direct measurement: This method uses physical battery properties, such as the voltage and impedance of the battery.

Book-keeping estimation: This method uses discharging current as the input and integrates the discharging current over time to calculate the SOC.

Adaptive systems: The adaptive systems are self-designing and can automatically adjust the SOC for different discharging conditions. Various new adaptive systems for SOC estimation have been developed.

Hybrid methods: The hybrid models benefit from the advantages of each SOC estimation method and allow a globally optimal estimation performance. The literature shows that the hybrid methods generally produce good estimation of SOC, compared to individual methods.

SOC Estimation Comparison

Extended Kalman Filter (EKF)

EKF SOC estimation is the process of estimating a battery's State of Charge (SOC) using the Extended Kalman Filter (EKF) method. It uses real-world measurements like voltage and current readings combined with a mathematical model of the battery that explains its electrical behavior.

Equations in EKF SOC estimation are:

Battery Model

A battery model is employed to predict how the battery's SOC changes over time. A simple SOC update equation might be:

- $SOC(t+1) = SOC(t) + \Delta SOC(t)$
- where $\Delta SOC(t)$ represents the change in SOC at each time step.

State Prediction

- During the prediction step, the battery model is used to estimate the SOC at the next time step:
- $SOC_predicted(t+1) = SOC(t) + \Delta SOC(t)$
- Here, $\Delta SOC(t)$ is often determined based on the battery's discharge and charge characteristics.

State Correction

- In the correction step, the predicted SOC is adjusted based on actual measurements, such as voltage and current readings:
- $SOC_{corrected}(t+1) = SOC_{predicted}(t+1) + K(t+1) * (Measured_SOC(t+1) - Predicted_SOC(t+1))$
- $Measured_SOC(t+1)$ is the SOC measured from real-world data.
- $Predicted_SOC(t+1)$ is the SOC predicted in the previous step.
- $K(t+1)$ is the Kalman Gain, calculated during the EKF process, which balances the prediction and measurement.

Battery Modelling

The longevity of batteries hinges on their inherent physical characteristics, and the aging process of battery cells is anything but straightforward. Merely minimizing power consumption at a given point in time won't elongate a battery's lifespan.

Instead, it's contingent on how power is used, the patterns of current extraction, and the levels of current involved. Consequently, creating a model to represent a battery's behavior becomes crucial when examining, estimating, or predicting its real-time performance.

Mathematical modeling of batteries holds significance for several reasons:

- **Efficient Battery Management Systems (BMS):** It aids in the development of efficient BMS, which are pivotal for optimizing a battery's usage.
- **Enhancing Charging and Discharging Techniques:** These models are instrumental in improving charging and discharging techniques, ultimately leading to enhanced battery capacity.
- **Capturing Power Consumption Influence:** They allow us to capture how power consumption affects the battery's performance accurately.
- **Preventing Damage:** Modeling helps prevent severe damage to batteries caused by overcharging or over-discharging, which can be detrimental.
- **Efficient Study of Battery Behavior:** It provides a faster and safer means to study how batteries behave under various operating conditions, contributing to better understanding.
- **Identifying Optimal Operating Limits:** These models help identify the operating limits that ensure the best possible lifetime for specific applications, thus aiding in optimal battery utilization.

Coulomb Counting Method

A battery's State of Charge (SOC) is estimated using the Coulomb counting method. Using this technique, the battery's discharge current is measured and integrated over time. The discharging current, $I(t)$, and the previously calculated SOC value, $SOC(t-1)$, are used to estimate the SOC at a specific time, represented as $SOC(t)$, using the formula below:

$$SOC(t) = SOC(t-1) + (I(t) / (Q_n * \Delta t))$$

It's crucial to remember that a number of variables, such as temperature, the battery's cycle life, the discharge current's size, and the battery's past performance, might affect how accurate the Coulomb counting approach is.

Coulomb Counting and EKF Combination

This technique improves SOC estimation accuracy by combining the Kalman filter method with the Coulomb counting methodology. The initial value used in Coulomb counting is refined using the Kalman filter in the Kalman + Ah approach, bringing it closer to the actual SOC value. For SOC estimate during prolonged operating times, Coulomb counting is then used. When compared to real values derived from discharge tests, this combination results in a noticeably lower SOC estimation error of 2.5 percent. On the other hand, an estimation error of 11.4% was much higher when the Coulomb counting approach was used alone.

Types of battery modelling methods

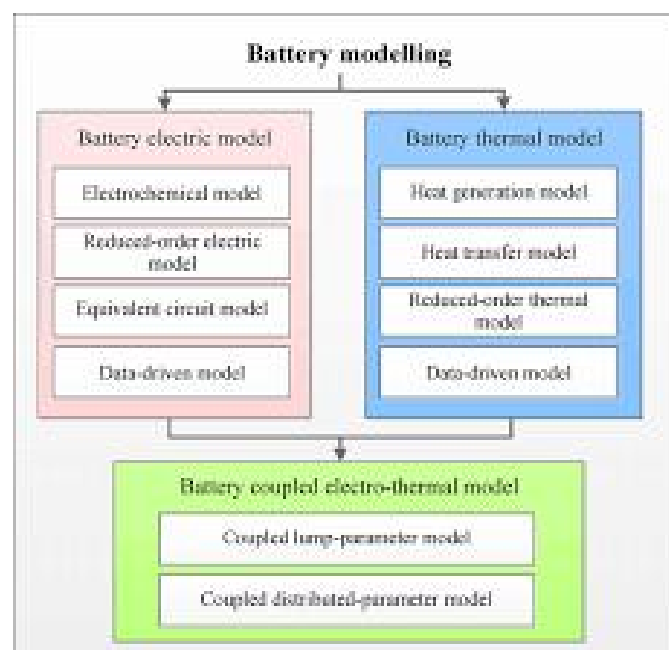


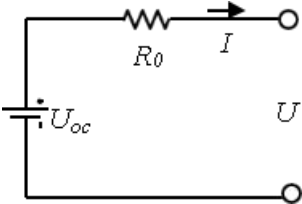
Figure 1: - Types of Battery Modelling methods

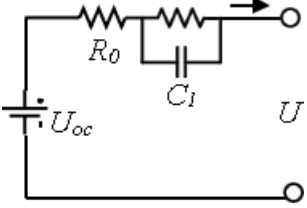
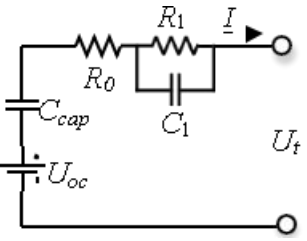
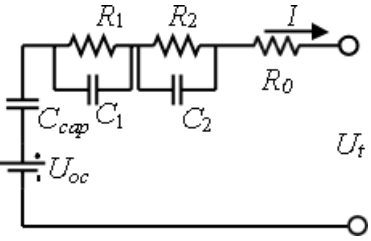
Equivalent Circuit Model

Resistance-Capacitance (RC) networks, an internal resistor, and a voltage source connected to the State of Charge (SOC) make up the Equivalent Circuit Model (ECM). Together, these elements explain the electrical relationship between different inputs, including temperature, SOC, and current, and the resulting terminal voltage. ECMs provide a more straightforward way to understand the electrical behavior of batteries than empirical models do. Additionally, ECMs give researchers a great deal of freedom in creating a structure that suits their particular applications.

In an ECM, a high-capacity capacitor or voltage source often reflects the battery's Open Circuit Voltage (OCV), whereas a resistor typically represents self-discharge. Aspects of the battery's behavior such as charge transfer, the double-layer effect within the electrode, and diffusion processes between the electrolyte and porous electrodes are represented by the RC pairs, each of which has a unique time constant. The Rint model, Thevenin model, Partnership for a New Generation of Vehicles (PNGV) model, and General ECM are the four widely utilized ECMs.

Non-Linear (GNL) model

Model	Equation
Rint Model 	The equation provided represents the terminal voltage, U_t, in a battery system. It's described as follows: $U_t = U_{oc} - I * R_0$. In this equation, U_t signifies the terminal voltage, U_{oc} represents the Open Circuit Voltage (OCV), I is the discharging current, and R_0 corresponds to the Ohmic resistance. This equation is commonly used in battery modeling to relate these variables in the context of battery behavior
Thevenin Model	The provided equation can be expressed as follows:

	$U_t = U_{oc} - U_1 - I \cdot R_0$. In this equation: • U_t represents the terminal voltage. • U_{oc} stands for the Open Circuit Voltage(OCV). • U_1 represents the voltage of the RC network. • I signifies the discharging current. • R_0 corresponds to the Ohmic resistance. • R_1 is the polarization resistance. • C_1 represents the polarization capacitance.
PNGV Model 	The equation provided can be expressed as: $U_t = U_{oc} - U_{cap} - U_1 - I \cdot R_0$. In this equation: • U_t represents the terminal voltage. • U_{oc} stands for the Open Circuit Voltage(OCV). • U_{cap} corresponds to the bulk capacitance. • U_1 represents the voltage of the RC network. • I signifies the discharging current. • R_0 is the Ohmic resistance.
GNL Model 	The provided equation can be expressed as: $U_t = U_{oc} - U_{cap} - U_1 - U_2 - I \cdot R_0$. In this equation: • U_t represents the terminal voltage. • U_{oc} stands for the Open Circuit Voltage(OCV). • U_{cap} corresponds to a voltage related to bulk capacitance. • U_1 represents a voltage related to one aspect of battery behavior. • U_2 represents another voltage related to battery behavior. • I signifies the discharging current. • R_0 is the Ohm resistance. • R_2 is the concentration polarization resistance. • C_2 is the concentration polarization capacitance.

CHAPTER 4

Known Data:

Sr.No.	Parameters	Value
1	Typical Capacity	31 Ah
2	Nominal Voltage	3.7 V
3	Maximum Current	62 A
4	Voltage	4.2 V
5	Continuous Current	155 A
6	Cut off Voltage	2.7 V
7	Charge	0 – 40° C
8	Energy Density	134 (Wh/kg)
9	Dimension	220 mm x 215 mm x 8.4 mm
10	Mass	0.86 kg
11	Discharge Operating Limits	-20°C to +60°C

COULOMB COUNTING METHOD

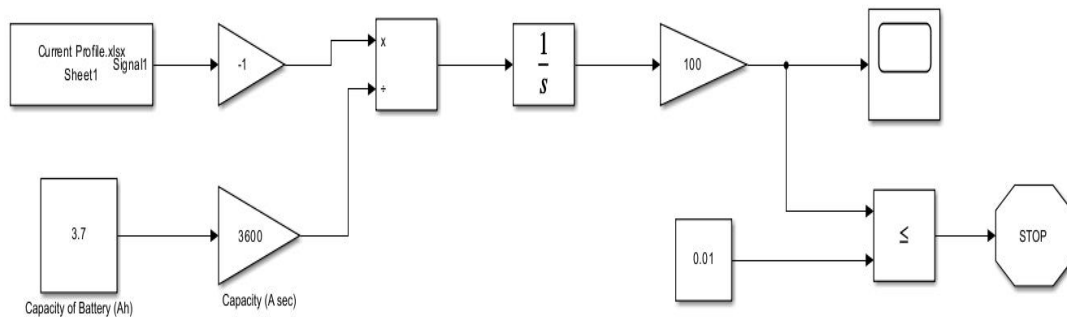


Figure 2: - Coulomb Counting Method in Simulink

KALMAN FILTER METHOD

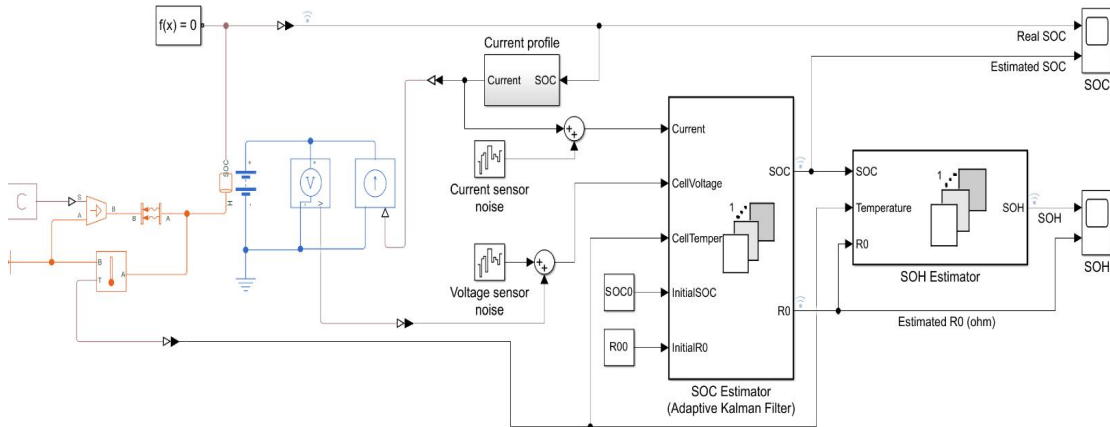


Figure 3: - Kalman Filter Method in Simulink

CHAPTER 5

OUTPUT PARAMETERS

MATLAB Code

```
%{
Parameters for Create a mathematical representation of the battery
total capacity of the mathematical battery
%}
M_B_CAPACITY_mAh = 2000; %IN mAh
M_B_CAPACITY_Ah = (M_B_CAPACITY_mAh/1000); %IN Ah
% Parameters for Battery State-of-Charge Estimation
%
% This example shows how to estimate the battery state-of-charge (SOC) by
% Using an Extended Kalman filter. The initial SOC of the battery is equal to
% 0.95.
% The estimator uses an initial condition for the SOC equal to 0.8. The
% Battery keeps charging and discharging for 6 hours. The extended Kalman
% Filter estimator converges to the real value of the SOC in less than 10
```

% Minutes and then follows the real SOC value. To use a different Kalman
 % Filter implementation, in the SOC Estimator (Kalman Filter) block, set
 % The Filter type parameter to the desired value.

%% System Parameters

SOC_LUT = [0, 0.1, 0.25, 0.5, 0.75, 0.9, 1]; % Vector of state-of-charge values, SOC

%T_LUT = [278, 293, 313]; % Vector of temperatures, T, (K)

TEMPRATURE_LUT = [278.15, 293.15, 313.15]; %temperatures, T, (K)

T_init_LUT = 293.15; %initial temperature

AH_LUT = 2; % Cell capacity, AH, (A*hr)

thermal_mass_LUT = 100; % Thermal mass (J/K)

cell_area = 0.1019; % Cell area (m²) h_conv = 5; % Heat transfer coefficient
 (W/(K*m²))

cell_mass = 0.720; %mass of the cell

cell_Cp_heat = 810.5326;

initialSOC_LUT = 0.95; % Battery initial SOC

V_NOMINAL_LUT = 3.7;

CAPACITY_LUT = [28.01, 27.63, 27.64]; % capacity look up to table data

C1_LUT = [1913.6, 12447, 30609;

4625.7, 18872, 32995;

23306, 40764, 47535;

10736, 18721, 26325;

18036, 33630, 48274;

12251, 18360, 26839;

9022.9, 23394, 30606];

V0_LUT = [3.49, 3.5, 3.51;

3.55, 3.57, 3.56;

3.62, 3.63, 3.64;

3.71, 3.71, 3.72;

3.91, 3.93, 3.94;

4.07, 4.08, 4.08;

4.19, 4.19, 4.19]; % Open-circuit voltage, V0(SOC, T), (V)

R0_LUT = [0.0117, 0.0085, 0.009;

0.011, 0.0085, 0.009;

0.0114, 0.0087, 0.0092;

0.0107, 0.0082, 0.0088;

0.0107, 0.0083, 0.0091;

0.0113, 0.0085, 0.0089;

0.0116, 0.0085, 0.0089]; % Terminal resistance, R0(SOC, T), (ohm)

R1_LUT = [0.0109, 0.0029, 0.0013;

0.0069, 0.0024, 0.0012;

0.0047, 0.0026, 0.0013;

0.0034, 0.0016, 0.001;

0.0033, 0.0023, 0.0014;

0.0033, 0.0018, 0.0011;

0.0028, 0.0017, 0.0011]; % First polarization resistance, R1(SOC, T), (ohm)

tau1_LUT = [20, 36, 39;

31, 45, 39;

109, 105, 61;

36, 29, 26;

59, 77, 67;

40, 33, 29;

25, 39, 33]; % First time constant, tau1(SOC, T), (s)

%% Kalman Filter

```

Q = [1e-4 0; 0 1e-4]; % Covariance of the process noise, Q
R = 0.7; % Covariance of the measurement noise, R
P0 = [1e-5 0; 0 1]; % Initial state error covariance, P0
SOC0 = 0.95; % Initial SOC for estimator
Ts = 1; % Sample time
%load('Battery_CELL_PERAMETERS_SIMULINK_FILE');
%%MATLAB_SIMULINK_FILE
%{
%%
Battery_CELL_PERAMETERS_SIMULINK_FILE.slx > this file contains both of
the
project objectives.
3> mathematical simulation file of the battery
2>SOC ESTIMATOR
3> BMS SIMULATION MODEL OF BMS SYSTEM
%}
%%
open_system('BMS_Modelling_project'); %to open the simulation file
sim ("BMS_Modelling_project")
figure(1)
surf(TEMPRATURE_LUT, SOC_LUT, R0_LUT); % Graphical visualization
title('Internal resistance(SOC, TEMP) Plot');xlabel('Temprature(Kelvin)');
ylabel('SOC');
zlabel('Internal resistance(Ohm)');
figure(2)
surf(TEMPRATURE_LUT,SOC_LUT,R1_LUT);
title('Resistance R1(SOC, TEMP) Plot');
xlabel('Temprature(Kelvin)');
ylabel('SOC');
zlabel('Internal resistance(Ohm)');
figure(3)
surf(TEMPRATURE_LUT,SOC_LUT,C1_LUT);
title('Capacitance C1(SOC, TEMP) Plot');
xlabel('Temprature(Kelvin)');
ylabel('SOC');
zlabel('Capacitance(mF)');
figure(4)
surf(TEMPRATURE_LUT,SOC_LUT,VO_LUT);
title('VO_LUT(SOC,TEMP) Plot');
xlabel('Temprature(Kelvin)');
ylabel('SOC');
zlabel('VO_LUT(volts)');

```

Simulink Model

An identical model of a lithium-ion battery cell and its matching Battery Management System (BMS) model are included in the constructed MATLAB model.

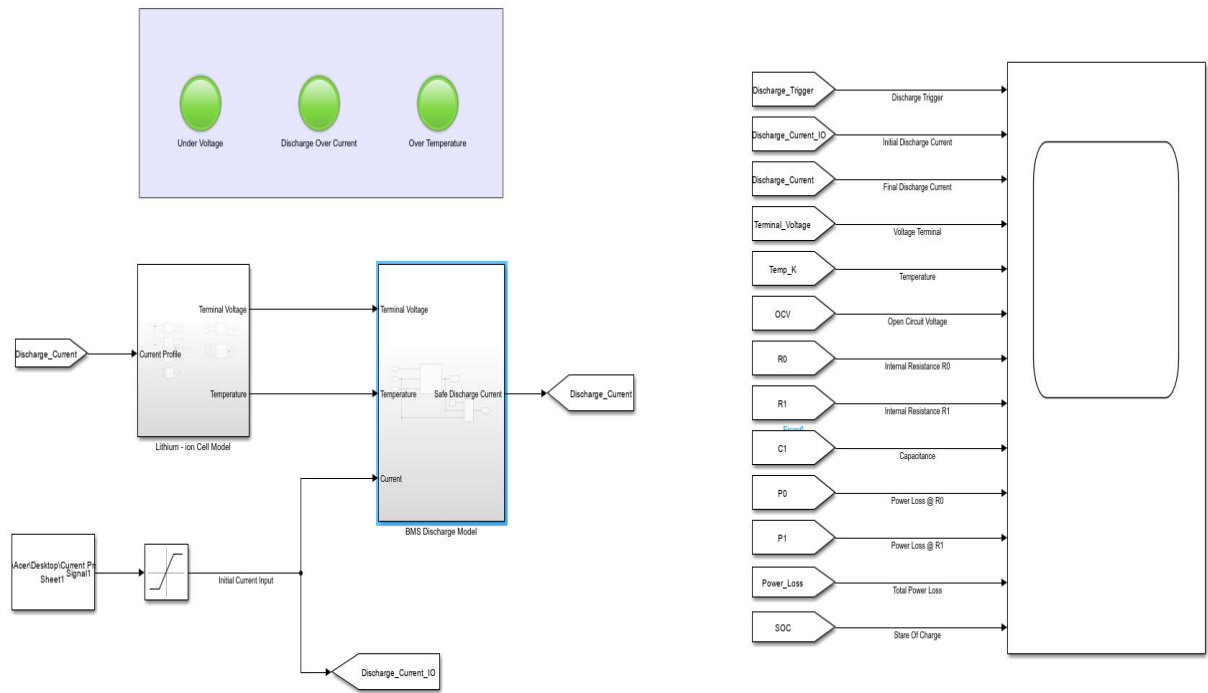


Figure 4: - Simulink Model of Equivalent Battery and BMS Setup

Components

1. Lithium-ion Model

We employed hybrid estimate for SOC estimation that combined the extended Kalman filter and coulomb counting.

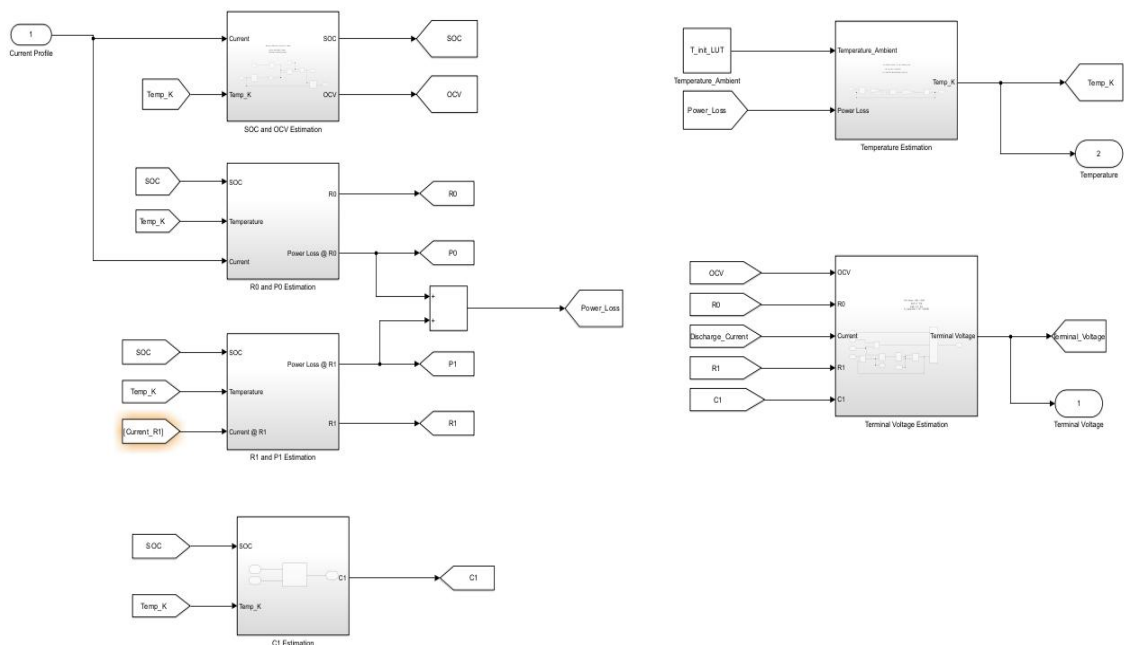


Figure 5: - Simulink Model of Equivalent Battery
SOC Estimation

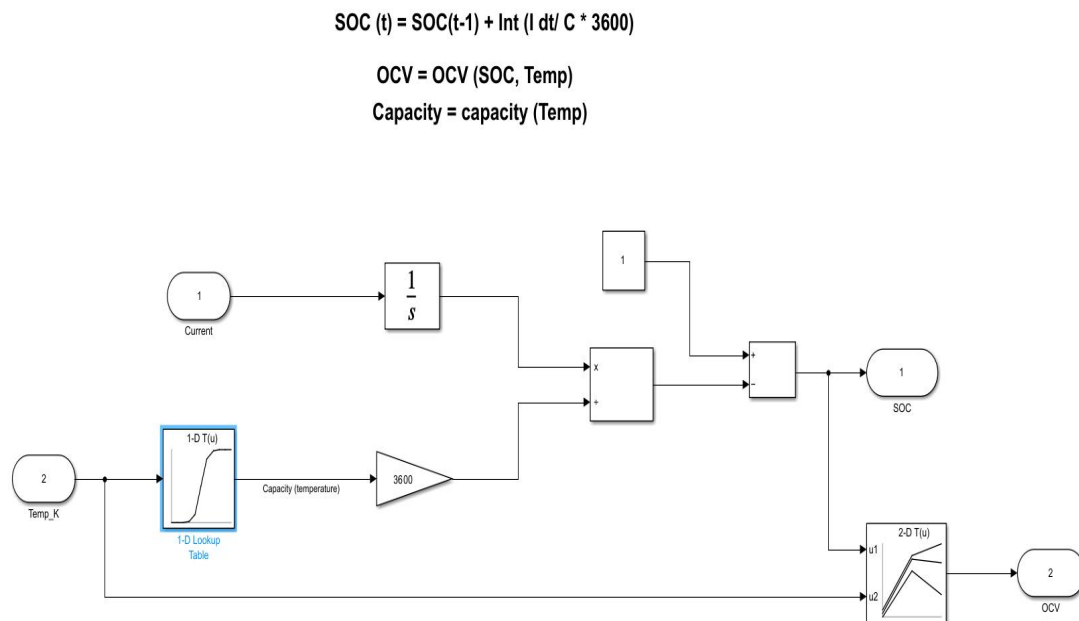


Figure 6: - SOC Estimation of Battery

1. R0

- R0 is often to as ohmic internal resistance, while the concept of battery internal resistance encompasses the combined resistance of all the resistive elements within the battery's equivalent circuit.
- $R0 = R0(\text{SOC}, \text{Temperature}) \rightarrow$ Lookup table.
- Voltage drop across R0($V1$) = $I * R0$
- Power Loss at R0($P0$) = Voltage drop * current

$$R0 = R0 (SOC, Temp)$$

$$\text{Voltage Drop} = I * R0$$

$$\text{Power Loss at } R0 = \text{Voltage Drop} * \text{Current}$$

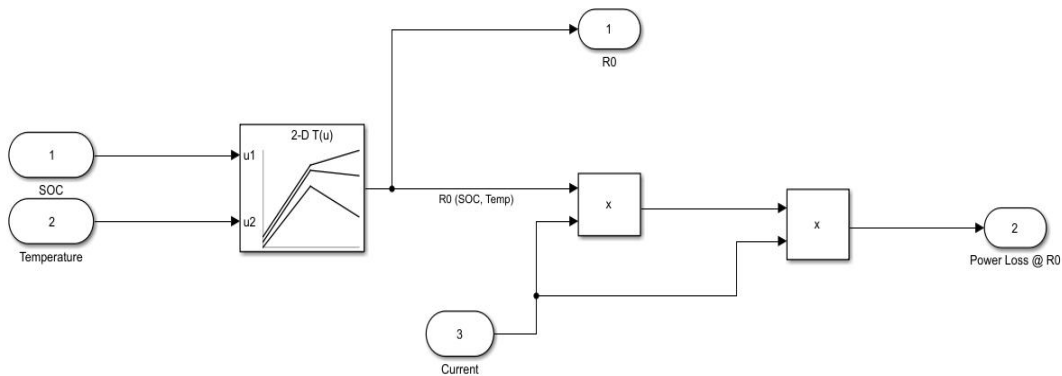


Figure 7: - R0 Subsystem

2. R1

• Polarization resistance (R1) can be understood as the resistance that rises at the interface between the electrodes and the electrolyte. Essentially, it represents an elevation in the resistance encountered by the electric current within a voltaic cell, and this increased resistance is primarily a consequence of chemical reactions occurring at the electrodes.

- $R1 = R1 (SOC, Temperature) \rightarrow$ Lookup table
- Voltage drop across R1 ($V1$) = $I * R1$
- Power Loss at R1 ($P1$) = Voltage drop * Current

$$R1 = R1(\text{SOC}, \text{Temp})$$

$$\text{Voltage Drop} = \text{Current across R1} * R1$$

$$\text{Power Loss at R1} = \text{Voltage Drop} * \text{Current at R1}$$

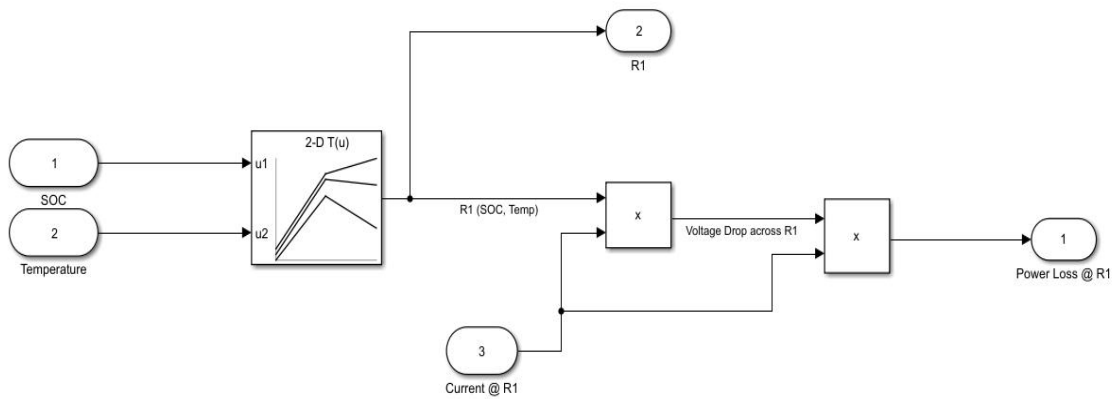


Figure 8: - R1 Subsystem

3. C1

- C1 is called as Polarization capacitance.
- $C1 = C1(\text{SOC}, \text{Temperature}) \rightarrow$ Lookup table.

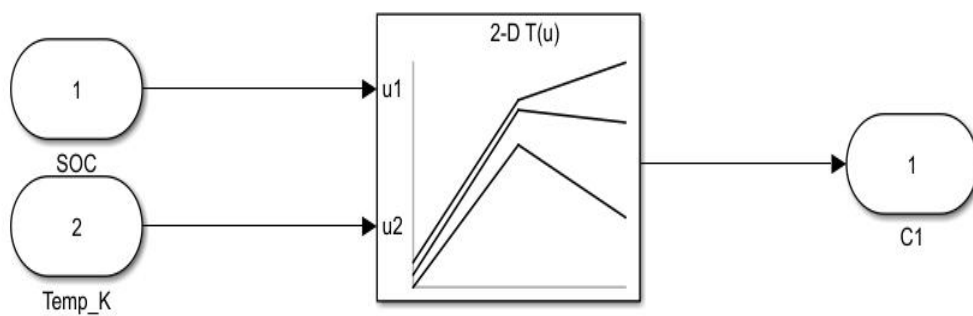


Figure 9: - C1 Subsystem

4. Temperature

We use a simplified representation of a thermal system equation,

$$C_t * dT/dt = (T_{amb} - T) * R_t + P_{loss}$$

$$R_t = h_{conv} * cell_area$$

C_t – Specific capacity (J/kg. K)

Here's a breakdown of the terms and their meanings:

- C_t : C_t stands for the specific heat capacity, which tell us how much energy is required to raise the temperature of a given amount of material by one degree Kelvin or Celsius (measured in J/kg. K) .
- dT/dt : This term represents the rate at which the temperature is changing with respect to time (measure in K/s). It indicates how quickly the system's temperature is increasing or decreasing.
- T_{amb} : T_{amb} signifies the ambient or surrounding temperature (measured in Kelvin or Degrees or Celsius). It represents the temperature of the environment where the system or component is located.
- T : T represents the temperature of the battery (measured in Kelvin or Degrees or Celsius).
- R_t : R_t represents the thermal resistance between battery and its surroundings. It's calculated by multiplying the convective heat transfer coefficient (h_{conv}), which characterizes how heat is transferred between the system and its surroundings, by the surface area (cell area) over which this heat transfer occurs. The unit of R_t is typically Kelvin per Watt (K/W).
- P_{loss} : P_{loss} signifies the rate at which heat is generated or power is lost within the battery (measured in Watts, W). This can result from various factors, including electrical resistance, chemical reactions, or other sources of heat generation.

$$C_t * dT/dt = (T_{amb} - T) * R_t + Power_Loss$$

$$R_t = H_{conv} * Cell\ Area$$

$$C_t = \text{Specific Heat Capacity (J/ Kg * K)}$$

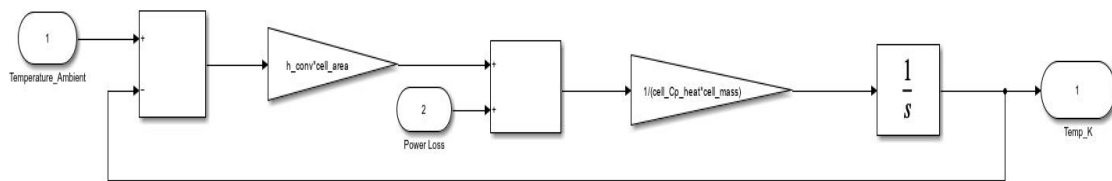


Figure 10: - Temperature Estimation Subsystem

5. Terminal Voltage

It can be determined by the following formula:

$$V_t = Vocv - Vr0 - Vr1$$

$$Vr0 = I * R0$$

$$Vr1 = I1 * R1$$

$$Vr1 = 1/C1 * \int I dt$$

- V_t : This represents the total voltage in the circuit or system.
- $Vocv$: This denotes the open-circuit voltage, which is the voltage measured across the circuit terminals when there is no current flowing. It's commonly known as the Thevenin Voltage (V_{th}).
- $Vr0$: This is the voltage associated with the ohmic or internal resistance ($R0$).
- $Vr1$: This signifies a voltage drop or component caused by another resistance ($R1$) in the circuit when current ($I1$) flows through it.

$$\begin{aligned}
 V_t &= V_{ocv} - V_{d1} - V_{d2} \\
 V_{d1} &= I * R_0 \\
 V_{d2} &= I * R_1 \\
 V_{capacitor} &= 1/C * \int I dt
 \end{aligned}$$

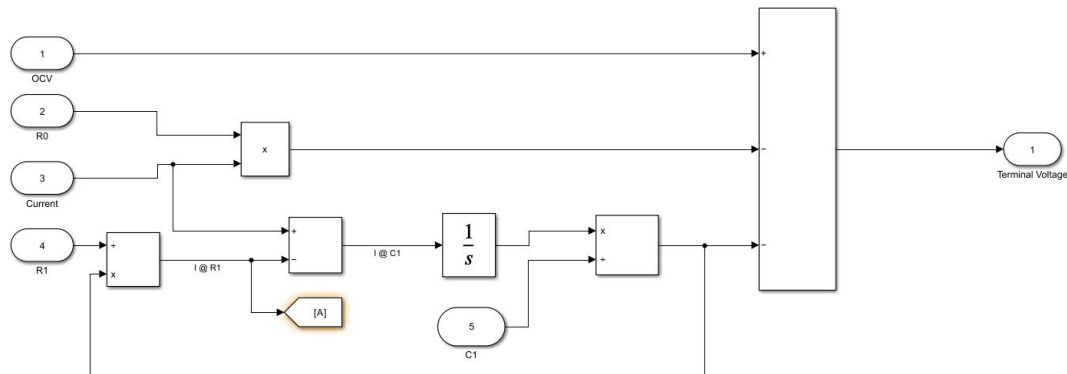


Figure 11: - Terminal Voltage Estimation Subsystem

6. BMS Model

In many battery-driven applications, such as electric cars, renewable energy storage, and portable electronics, a battery management system (BMS) is an essential part. Its main responsibility is to guarantee the battery pack operates safely and effectively. The following are some characteristics of the BMS Model:

1. Under Voltage Protection (UVP):

- o UVP keeps the battery from draining to a point where the cells or their functionality could be harmed.

- o To stop additional discharge, the BMS can either switch off the load or sound an alert when the battery voltage falls below a predetermined threshold.

2. Overcurrent Protection (OCP):

- o OCP protects the battery pack from excessive current flow, which can happen as a result of overloading or short circuits.

- o The BMS continuously checks the current and, if it exceeds a preset limit, can cut the current or disconnect the battery.

3. Overheating Protection (Thermal Management)

To keep the battery from running at temperatures that could harm the cells or create safety risks, overheating protection is essential.

o To maintain acceptable temperature levels, the BMS may incorporate temperature sensors inside the battery pack and, if necessary, turn on heating or cooling systems.

· To stop thermal runaway in severe situations, the BMS has the ability to disconnect the battery if temperatures reach a critical point.

Logic:

The discharge current will be 0 in the event that any of the following circumstances are met (open circuit).

- a. Terminal Voltage is less than 2.2 V which is cut off voltage.
- b. Maximum Discharge current is more than 20A
- c. Temperature limit (-20 ° C + 60 °C).

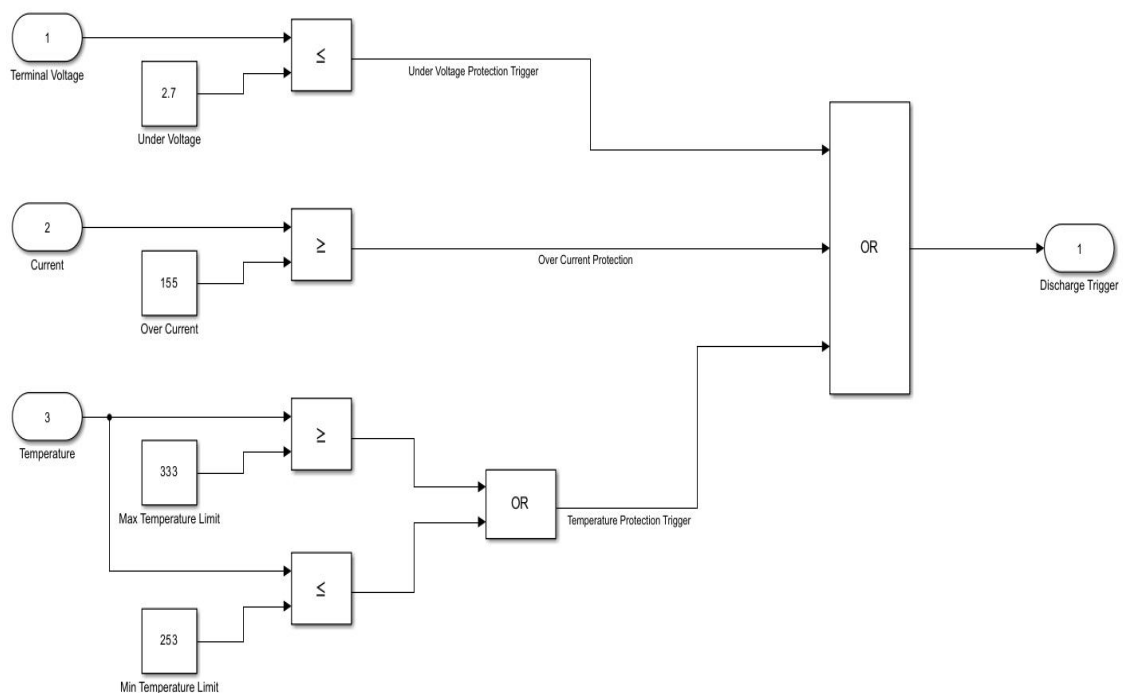


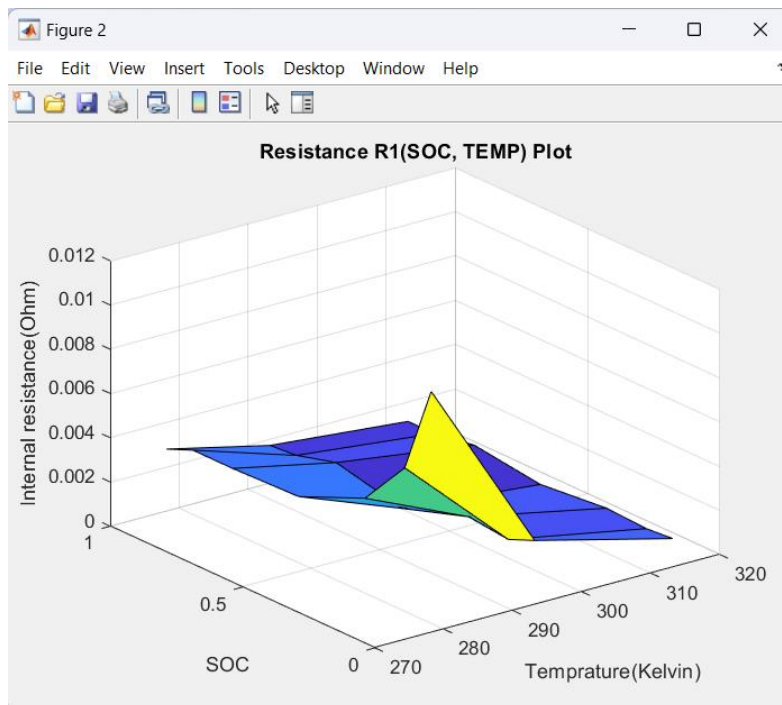
Figure 12: - BMS Subsystem

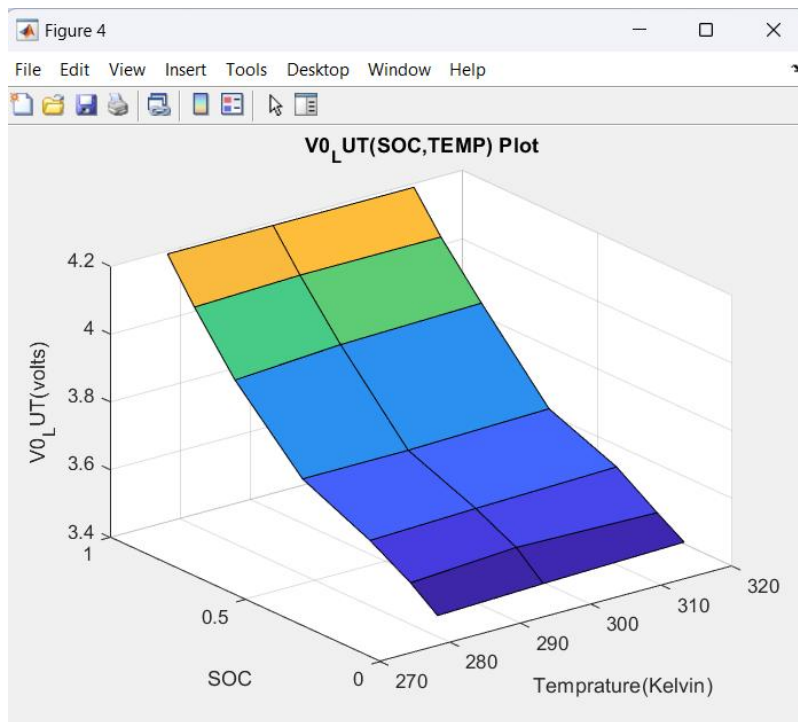
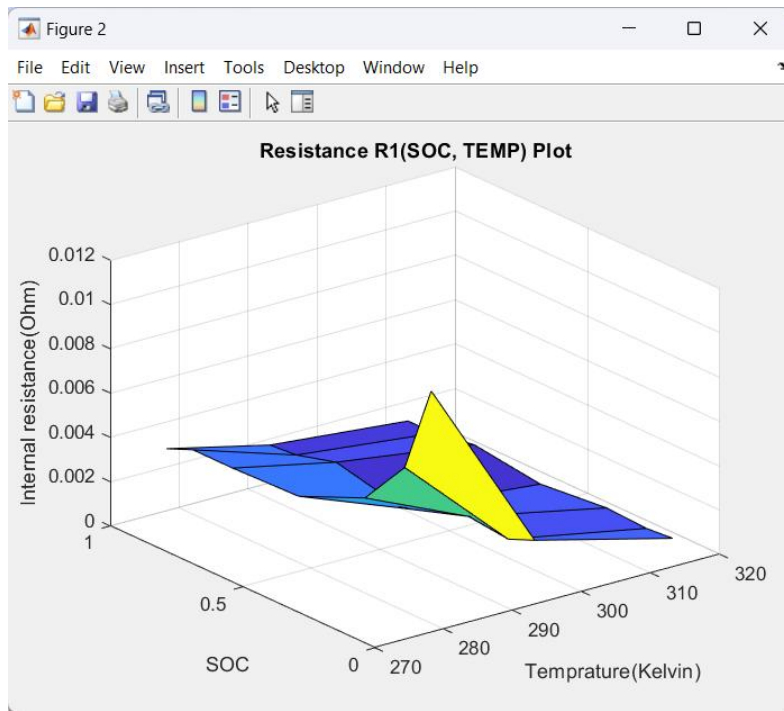
CHAPTER 6

RESULT

1. MATLAB Plot Results

The data that is displayed includes charts that show how the battery's temperature and State of Charge (SOC) affect the values of R_1 (polarization resistance), R_0 (ohmic or internal resistance), C_1 (capacitance), and E_m (open circuit voltage). These charts show how these characteristics alter in response to changes in the battery's state of charge and temperature.





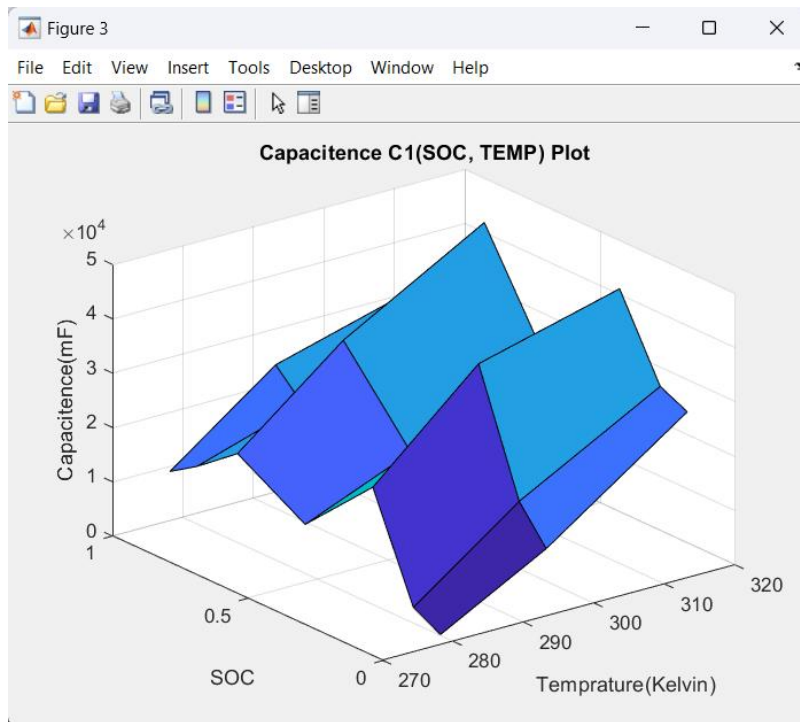


Figure 13: - MATLAB Plot Results

2. Simulation Results

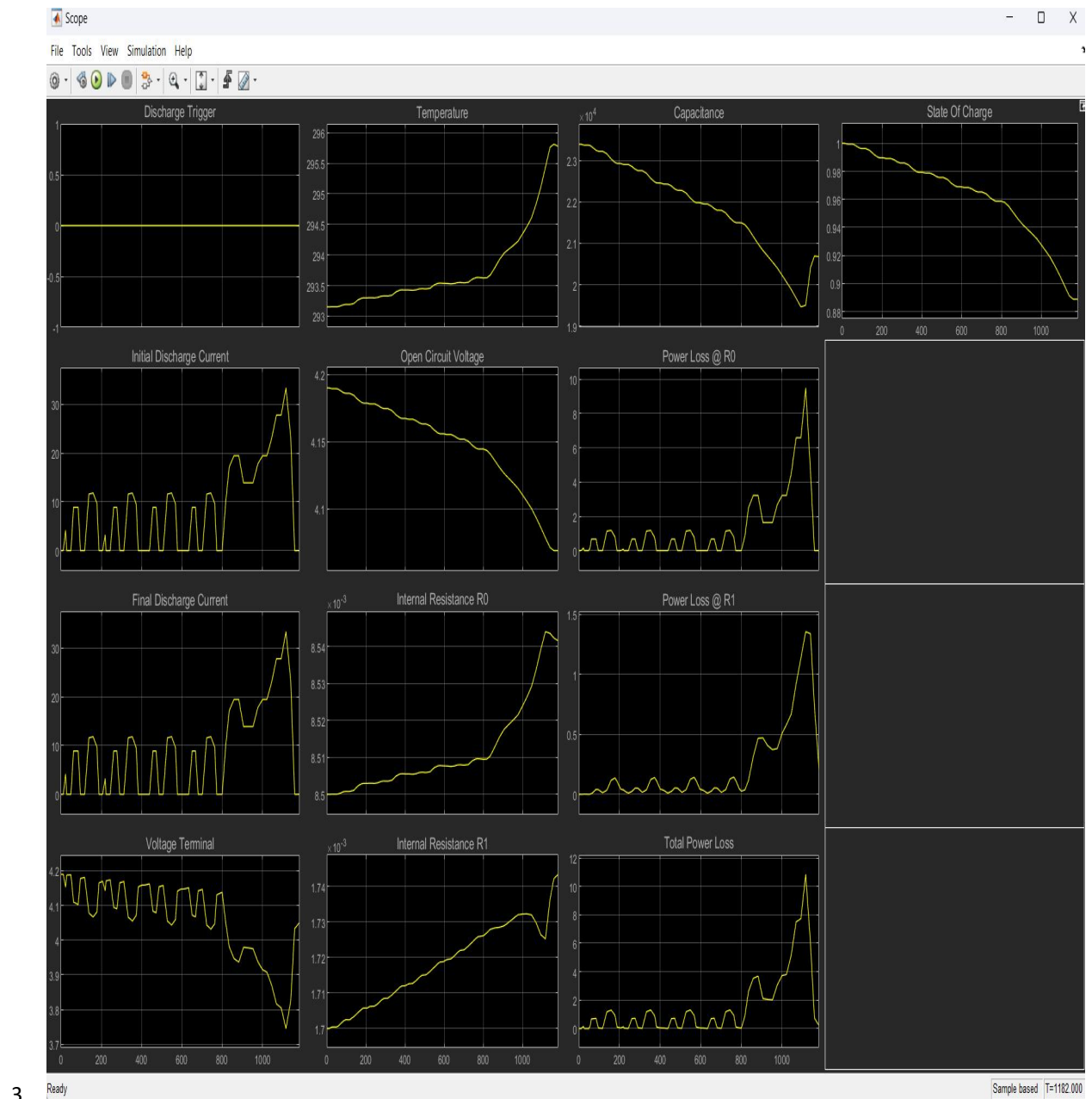


Figure 14: - Scope of BMS Model

CHAPTER 7

OBSERVATION AND CONCLUSION

To sum up, the goal of this project is to use MATLAB to create a thorough Battery Management System (BMS) model, with a primary emphasis on precisely calculating a battery pack's State of Charge (SOC) during cycles of charging and discharging.

• In order to simulate how the battery would react to different loads and charge/discharge scenarios, the project entails developing a mathematical representation of the battery's behavior that takes internal resistance, discharge capacity curves, and voltage-current characteristics into account. With the use of accessible measurements, such as voltage and current readings, many SOC estimating methods, including Coulomb counting, Extended Kalman Filtering, and Equivalent Circuit Models, will be investigated and evaluated for accuracy.

• Using hardware interfaces for real-time data collection or preexisting datasets, data acquisition will be carried out to confirm the BMS model's performance, contingent on resource availability. Furthermore, the project will generate graphical outputs that allow for a comparison of estimated and actual SOC values by displaying the performance of SOC estimates over time.

• In essence, this project aims to address the critical requirement for precise state-of-charge (SOC) estimation in battery management, which is necessary for the secure and efficient operation of applications such as electric vehicles and renewable energy systems. Through the development and validation of a robust BMS model, this initiative consistently advances battery technology and its everyday integration.

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With the submission of this project report, I acknowledge that I have not copied the work/results/or text from other sources. All the used work from other sources is cited properly with the right format. In case the plagiarism is more than 20%, the instructor has the authority to cancel my project submission.